

IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired

R26-IT-134

Project Proposal Report

Sandaruwan U D H – IT22074690

B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

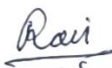
February 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Sandaruwan U D H	IT22074690	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Mr. Supunya Swarnakantha	12/06/2024	

ABSTRACT

Indoor navigation remains a significant challenge for visually impaired individuals due to the unavailability of GPS signals in enclosed environments and the limitations of traditional assistive tools such as white canes, which detect only ground-level obstacles. Although recent studies have introduced camera-based smart glasses using deep learning for object recognition, many existing systems still lack depth-aware perception, synchronized multi-sensor integration, and compact hardware designs optimized for real-time indoor safety. This creates a research gap in developing an efficient, wearable, and multi-sensory navigation device that improves hazard detection reliability while maintaining portability and affordability.

This research proposes an IoT-based smart spectacle designed specifically for indoor navigation and safety enhancement of visually impaired users. The system integrates a camera, LiDAR sensor, and Inertial Measurement Unit (IMU) within a compact wearable frame, enabling multi-sensory data fusion for accurate obstacle detection, distance estimation, and staircase identification. A Raspberry Pi module performs real-time edge processing using deep learning-based object detection and depth mapping techniques. The system provides immediate audio feedback to inform users about obstacle type and distance, ensuring timely hazard awareness.

The expected outcome is a lightweight, cost-effective, and scalable assistive device capable of operating offline with improved detection accuracy compared to single-sensor systems. Socially, the solution aims to enhance independence, mobility confidence, and safety for visually impaired individuals. Commercially, it offers potential for scalable production in developing regions, contributing to inclusive technology development and smart assistive IoT innovations.

Keywords: IoT; Smart Spectacle; Indoor Navigation; Multi-Sensor Fusion; Assistive Technology

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor and co-supervisor for their continuous guidance, valuable feedback, and academic support throughout the development of this research project titled “IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired.” Their insights and encouragement greatly contributed to shaping the technical direction and research quality of my component **Smart Spectacle Hardware Module with Multi-Sensory Integration**.

I am also thankful to the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment, laboratory facilities, and resources necessary to carry out this research under the IT4010 Research Project module.

My appreciation extends to my fellow group members for their collaboration, knowledge sharing, and teamwork in integrating hardware, AI, and navigation components into a unified assistive system. Their cooperation made the research process productive and efficient.

Finally, I would like to thank my family and friends for their constant encouragement, understanding, and moral support throughout the project period. Their motivation has been invaluable in completing this research successfully.

Table of Contents

ABSTRACT.....	3
ACKNOWLEDGEMENT.....	4
LIST OF TABLES.....	7
LIST OF FIGURES	8
LIST OF ABBREVIATIONS	9
LIST OF APPENDICES.....	10
1. INTRODUCTION.....	11
1.1 Background and Context.....	12
1.2 Literature Review.....	14
1.3 Research Gap	16
2. OBJECTIVE	18
2.1 Main Objective.....	18
2.2 Specific Objectives	18
3. METHODOLOGY	19
3.1 Research Approach	19
3.2 Technical Methods and Algorithms.....	19
3.3 Tools and Platforms with Justification.....	20
3.4 Data Collection and Ethical Considerations	21
3.5 Validation Strategy and Performance Metrics	22
4. HIGH-LEVEL SYSTEM ARCHITECTURE	23
4.1 Overview of the Proposed Solution	23
4.2 System Components.....	23
4.3 High-Level System Diagram (Components).....	24
4.4 Highlight: Individual Component and Its Parts (Smart Spectacle Hardware Module).....	25
4.5 Component Interactions	25
4.6 Data Flow Explanation	26
4.7 Scalability and Security Considerations	27
5. USER REQUIREMENTS.....	28
5.1 Stakeholder Identification	28
5.2 Functional Requirements	28
5.3 Non-Functional Requirements	29
6. COMMERCIALIZATION PLAN	30
6.1 Target Market and Customer Persona.....	30

6.2 Value Proposition	30
6.3 Revenue Model	31
6.4 Cost Estimation.....	31
6.5 Pricing Strategy.....	31
6.6 Competitive Advantage	32
6.7 Intellectual Property Considerations.....	32
7. BUDGET AND JUSTIFICATION	33
7.1 Estimated Budget Breakdown.....	33
7.2 Hardware Cost Justification	33
7.3 Software Requirements	34
7.4 Cloud Services	34
7.5 Human Effort Cost.....	35
8. WORK BREAKDOWN STRUCTURE (WBS)	36
8.1 Task Breakdown per Member.....	36
8.2 Timeline Alignment	37
8.3 Balanced Workload Distribution	38
9. GANTT CHART	39
10. REFERENCES LIST.....	40
11. APPENDICES	42
Appendix A - Survey Instruments	42
Appendix B - Detailed System Diagrams	43
Appendix C – Preliminary UI Mockups	44
Appendix D – Risk Analysis.....	45
Appendix E – Ethical Considerations	46

LIST OF TABLES

Table 1: Value Proposition of the Proposed Smart Spectacle System	30
Table 2 : Estimated Prototype Cost of the Proposed Smart Spectacle System.....	31
Table 3 : Estimated Budget Breakdown for Prototype Development.....	33
Table 4 : Software Tools Used in the Proposed System.....	34
Table 5 : Task Breakdown per Member for the Proposed Smart Spectacle System.....	37
Table 6 : Risk Analysis for the Smart Spectacle Hardware Module (Component 3)	45

LIST OF FIGURES

Figure 1 : High-Level System Architecture	24
Figure 2 : Data Flow of the Proposed Smart Spectacle System	26
Figure 3 : Use Case Diagram	29
Figure 4 : Intellectual Property Protection Framework	32
Figure 5 : Gantt Chart for the Proposed Smart Spectacle Research Project	39
Figure 6 : Smart Spectacle Hardware Architecture Diagram	43
Figure 7 : Sensor Data Processing Pipeline	44
Figure 8 : Preliminary User Interface Mockups for the Guardian Monitoring Mobile Application	44

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
BLE	Bluetooth Low Energy
CPU	Central Processing Unit
GPS	Global Positioning System
ICT	Information and Communication Technology
IMU	Inertial Measurement Unit
IoT	Internet of Things
LiDAR	Light Detection and Ranging
OpenCV	Open Source Computer Vision Library
SDG	Sustainable Development Goals
SOS	Save Our Souls (Emergency Distress Signal)
TFLite	TensorFlow Lite
UWB	Ultra-Wideband
Wi-Fi	Wireless Fidelity
YOLO	You Only Look Once (Object Detection Algorithm)

LIST OF APPENDICES

Appendix	Title	Page No.
Appendix - A	Survey Instruments	41 - 42
Appendix - B	Detailed System Diagrams	42 - 43
Appendix - C	Preliminary UI Mockups	43
Appendix - D	Risk Analysis	44
Appendix - E	Ethical Considerations	45

1. INTRODUCTION

Visual impairment also affects the level of independent mobility, especially for individuals with visual impairment, as it becomes difficult for them to navigate through indoor environments, especially due to the unavailability and unreliability of GPS systems. According to the World Health Organization, millions of people worldwide experience moderate to severe vision impairment [1], which becomes a significant obstacle for them when navigating indoor environments. Indoor environments also consist of unpredictable obstacles, which become a major source of injury for individuals with visual impairment [2], [3].

Traditionally, mobility devices such as a white cane and a guide dog are found to be very helpful to a visually impaired person, although there are certain limitations to each of these devices [2]. For example, a white cane can only assist a visually impaired person in identifying obstacles on the floor, but it cannot assist in identifying a beam hanging from the ceiling or a staircase extending from the floor [2]. Similarly, the guide dog requires a lot of training and maintenance [3]. However, with the advent of embedded systems, IoT, and edge-based artificial intelligence, the concept of a smart device has been introduced.

Recently, various studies have proposed smart glasses with cameras, ultrasonic sensors, or LiDAR sensors to recognize obstacles and provide audio feedback [4], [5], [6], [7], [8]. However, existing approaches are limited by their bulky nature, poor sensor integration, dependency on cloud environments, or limited depth perception capabilities [4], [9], [7]. Such constraints are particularly problematic for developing countries where internet availability is poor.

This research proposes a Smart Spectacle Hardware Module with Multi-Sensory Integration as a component of a broader project: “IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired.” The proposed module is intended to provide a compact, lightweight, and modular wearable hardware architecture that integrates a camera, a LiDAR sensor, and an Inertial Measurement Unit (IMU) along with a Raspberry Pi edge computing platform [10].

This proposed hardware module is expected to improve the reliability of the entire system, staircase detection capabilities, and safety of the visually impaired. The proposed system is intended to provide enhanced freedom of movement and confidence for the visually impaired population while being affordable enough for real-world deployment scenarios.

1.1 Background and Context

Visual impairment is a major public health challenge in the world, and this has a major implication for independent and safe movement. Individuals with visual impairment face significant challenges in moving around indoor environments, where GPS technology is not reliable or available. According to the World Health Organization (WHO), millions of people worldwide are living with moderate and severe visual impairment [1], and this is a major challenge to their independent movement. The inability to recognize obstacles such as staircases, hanging objects, furniture, and corridors is a major challenge to people living with visual impairment [2], [3].

The effects of this problem have been observed in developing countries because they lack assistive technology due to high costs. In developing countries like Sri Lanka, there is a lack of infrastructure for accessibility, indoor navigation technology, and internet connectivity [3]. The independence of visually impaired individuals to move in different environments makes the development of assistive technology both socially and technologically important.

Traditionally, mobility assistance has been provided through devices such as the white cane and guide dogs. Although these devices are very useful, they have a number of drawbacks. For instance, the white cane can detect obstacles on the ground but may not detect obstacles above eye level or stairs leading downwards [2]. On the other hand, guide dogs require very expensive and time-consuming training and may not be accessible to everyone [2].

Recent advancements in embedded systems, IoT, and edge-based artificial intelligence have led to the development of smart wearable devices. Several studies have proposed smart glasses equipped with cameras, ultrasonic sensors, and LiDAR sensors to perform obstacle detection and provide real-time feedback [4], [5], [6], [7], [8]. Deep learning-based object detection frameworks such as YOLO have enabled real-time recognition of environmental objects with high accuracy [11]. Additionally, wearable navigation devices utilizing semantic SLAM techniques have demonstrated improved spatial awareness and navigation assistance [7].

However, despite these developments, existing solutions are still found to be plagued by many limitations and disadvantages. For instance, some of them are cumbersome and not comfortable for long-term usage. Others are found to be heavily dependent on cloud computing and internet connectivity for processing information [4], [9], [7]. Furthermore, limited integration of sensor fusion also makes them unreliable in complex indoor situations [5], [8].

To mitigate these issues and challenges, this research is proposing a Smart Spectacle Hardware Module with Multi-Sensory Integration as part of its larger project titled “IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired.” The proposed Smart Spectacle Hardware will integrate a camera, LiDAR sensor, and Inertial Measurement Unit with a Raspberry Pi edge computing platform [10]. This will reduce its dependency on cloud computing and provide fast responses at the same time. The modular design will also ensure its compactness and lightness for practical usage.

1.2 Literature Review

Recent research on assistive indoor navigation for blind and visually impaired (BVI) users has expanded from traditional aids toward wearable, sensor-rich, and AI-supported systems. A key trend is the shift from single-sensor solutions (e.g., ultrasonic only) to multimodal perception using combinations of cameras, depth sensors (LiDAR/RGB-D), and inertial sensors, often paired with embedded or edge computing [2], [3].

- **Wearable navigation and perception approaches**

Wearable vision + mapping/localization is a significant area. Chen et al. presented a wearable device using semantic visual SLAM and an RGB-D camera [7]. Surveys also highlight the fact that the indoor navigation solution should meet the needs of BVI users, such as reliability, cognitive load, and robustness [3]. Plikynas et al. presented a review of state-of-the-art indoor navigation technologies for visually impaired users and mapped them to the needs of the user [3].

- **Embedded/edge processing and real-time feedback**

Edge computing is increasingly used to reduce cloud dependency and improve response time. For example, a Raspberry Pi-based wearable navigation support system has been proposed with multi-sensor input and haptic/voice interfaces, demonstrating feasibility of onboard processing [4]. Kumar et al. presented a Raspberry Pi-based real-time object recognition system with voice feedback for visually impaired users, supporting the practicality of embedded vision approaches [10]. Deep learning frameworks remain a common foundation for real-time detection; YOLO-based object detection approaches are widely used due to speed and accuracy in real-time contexts [11].

- **Smart glasses designs and multi-function wearables**

Recent smart-glasses research explores combining perception and user feedback through audio/tactile outputs. Bastola et al. presented design and development of multi-functional glasses for blind and visually impaired users, reflecting ongoing work on wearable form-factor feasibility and feature integration [4]. A wearable assistive system combining object detection, distance measurement, and tactile presentation demonstrates that multi-sensory feedback can enhance situational awareness, though it may introduce hardware bulk and complexity [8]. Recent reviews also summarize computer vision and sensor-based indoor navigation methods with auditory guidance strategies [3], [9]. SLAM-focused surveys highlight continued challenges in real-time localization accuracy, semantic understanding, and robustness under indoor lighting/texture ambiguity [7].

- **Existing products and services**

However, such commercial solutions also show some of the limitations of such systems. For instance, OrCam MyEye is a wearable device intended for text reading and object/product recognition but is often beyond people's budgets and is not specifically suited for depth-aware indoor navigation and staircase recognition. Envision is focused on assistive access of visual information such as reading and recognition through smart glasses-related solutions but is not specifically suited for navigation-related depth sensing and indoor safety. Aira is a service focused on human assistance through video streaming but is highly dependent on availability and connectivity.

1.3 Research Gap

Although there has been significant progress in the development of assistive technologies for visually impaired persons, there are still some limitations in the existing indoor navigation systems for visually impaired persons. Existing research studies and systems are mostly concentrated on specific aspects of providing navigation assistance, and there is still a lack of a comprehensive, compact, and reliable solution for real-world indoor environments.

Based on the literature review, the following research gaps have been identified:

- **Limited sensor integration:**

Many existing systems rely on a single sensor such as a camera or ultrasonic sensor for obstacle detection. This limits their ability to accurately detect different types of obstacles such as overhead objects, stairs, or elevation changes [3], [6].

- **Bulky hardware designs:**

Several wearable navigation devices proposed in previous research use complex sensor setups and large hardware modules, making them less practical for everyday wearable use [7].

- **Dependence on cloud processing:**

Some assistive systems rely on cloud computing for image processing and object recognition. This introduces latency and requires stable internet connectivity, which is not always available in indoor environments [4], [9].

- **Limited depth perception capability:**

Vision-based systems without dedicated depth sensors often struggle to detect changes in elevation such as staircases or downward steps, which are critical safety hazards for visually impaired users [5].

- **Insufficient multi-sensor fusion:**

Existing smart glasses solutions often focus on object recognition but lack effective integration between camera systems, motion sensors, and distance sensors to provide reliable navigation support [8], [10].

Due to these limitations, there is a clear need for a compact and integrated wearable navigation solution that combines computer vision, depth sensing, and motion tracking within a single system.

Therefore, this research proposes a **Smart Spectacle Hardware Module with Multi-Sensory Integration**, which combines a camera module, LiDAR distance sensor, and IMU motion sensor with a Raspberry Pi edge computing platform. By performing real-time processing on an embedded device, the proposed system aims to improve obstacle detection, staircase recognition, and navigation safety while reducing dependency on cloud-based services.

The proposed solution contributes to the field of assistive technology by introducing a lightweight, cost-effective, and wearable navigation system designed specifically to improve independent mobility for visually impaired individuals.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop an assistive smart spectacle system that improves safe and independent indoor navigation for visually impaired individuals, thereby enhancing their mobility, confidence, and quality of daily life.

2.2 Specific Objectives

- To design and develop a wearable smart spectacle hardware platform capable of supporting multiple sensors for indoor navigation assistance.
- To integrate multiple sensors like a camera module, LiDAR sensor, and IMU sensor to detect environment information and motion information to detect obstacles and assist in navigation.
- To develop a real-time obstacle detection system using computer vision techniques to detect obstacles and assist in navigation.
- To develop a sensor fusion mechanism that combines visual data, distance measurements, and motion information to improve navigation accuracy.
- To develop an audio feedback mechanism to assist visually impaired persons with navigation.
- To evaluate the performance of the proposed system in detecting obstacles and assisting indoor navigation for visually impaired individuals.

3. METHODOLOGY

3.1 Research Approach

This research is conducted under a research approach known as Design Science Research, where the main aim is to design and develop a technological artifact that addresses a real-world problem. The technological artifact that is being proposed in this research is a Smart Spectacle Hardware Module with Multi-Sensory Integration, which is aimed at helping visually impaired persons navigate indoor environments.

In addition, this research has adopted an experimental evaluation approach, where the effectiveness of the proposed system will be evaluated. The prototype that is being developed in this research will be used in controlled indoor environments to test its effectiveness in detecting obstacles and recognizing objects.

3.2 Technical Methods and Algorithms

The proposed system uses various techniques for detecting hazards in the environment and for navigation.

The camera module captures real-time visual data from the surrounding environment. Computer vision techniques and deep learning models such as **YOLO (You Only Look Once)** will be used for object detection and recognition.

A **LiDAR distance sensor (VL53L1X)** is used to measure the distance between the user and surrounding obstacles. This helps detect objects, walls, and elevation changes such as staircases.

The **Inertial Measurement Unit (IMU) sensor (MPU6050)** is used to calculate the orientation of the user's head. It can also detect the movement of the user's head. It can be used for navigation and can increase the overall accuracy of the system.

The sensor data is processed using a **Raspberry Pi computing platform** for detecting hazards in real-time.

The proposed system provides an **audio-based feedback mechanism using earphones** for user awareness about hazards in their surroundings.

3.3 Tools and Platforms with Justification

The development of the proposed system requires several hardware and software tools.

- **Raspberry Pi Zero 2 W**

The Raspberry Pi Zero 2 W will be used as a processing unit for the proposed system. This is because it is a small-sized, low-power-consuming, and edge computing-capable unit, which is appropriate for a wearable device.

- **Raspberry Pi Camera Module V2**

The Raspberry Pi Camera Module V2 will be used to acquire real-time visual data from the surrounding environment. The camera captures images that are used for object detection and environmental analysis.

- **VL53L1X LiDAR Sensor**

The VL53L1X LiDAR Sensor will be used for distance and obstacle detection. This is an essential feature for recognizing objects and elevation changes such as staircases.

- **MPU6050 IMU Sensor**

The MPU6050 IMU Sensor will be used to detect head orientation and movement. This improves navigation by using motion sensors.

- **Rechargeable LiPo Battery**

A 3.7V 3000mAh Li-Polymer battery (454487) will be used as the primary power source for the wearable device. Since the Raspberry Pi requires a stable 5V power supply, an MT3608 boost converter will be used to step up the battery voltage from 3.7V to 5V.

This power configuration allows the system to operate as a portable and rechargeable wearable device.

- **Software Tools (Python and OpenCV)**

For software development, Python programming language and OpenCV libraries will be used.

3.4 Data Collection and Ethical Considerations

The data collection for this research will be on objects found in an indoor environment and navigation hazards that a visually impaired person may encounter. The data will be collected from an indoor environment such as a hallway, room, or staircase.

The data to be used in the research includes objects like a chair, table, door, walls, staircase, and other obstacles found in an indoor environment. The data collected in the research environment will be used to train the object detection model used in the system.

The data used in the research may be images collected during experimentation or data used by other researchers in computer vision research. The data collected in the research environment will be used to train the object detection model to detect objects found in an indoor environment.

Ethical Considerations,

The research does not involve obtaining personal or sensitive data from individuals. When obtaining any form of image data in the experiments, the data will only include environmental objects and obstacles. When people are present in the images taken in the environment, their identity is not stored.

The experiments are carried out in a manner that respects privacy and safety while adhering to ethical guidelines in carrying out research.

3.5 Validation Strategy and Performance Metrics

The performance of the proposed system will be evaluated through experimental testing in indoor environments.

The following performance metrics will be used to evaluate the system:

- **Obstacle detection accuracy** - ability of the system to correctly detect obstacles in the environment.
- **Object recognition accuracy** - accuracy of the computer vision model in identifying objects.
- **Distance measurement accuracy** - precision of the LiDAR sensor in detecting obstacle distance.
- **Response time** - time taken for the system to detect hazards and generate audio alerts.
- **Navigation assistance reliability** - effectiveness of the system in assisting users to safely navigate indoor spaces.

The experimental results will be analyzed to determine whether the proposed system improves safety and navigation support for visually impaired individuals.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed solution is an **IoT-based smart spectacle** that assists visually impaired users with indoor navigation and safety. The system collects environmental and motion data using multiple sensors, processes the data on an **edge computing unit (Raspberry Pi)**, and provides **audio feedback** to the user through an earphone.

4.2 System Components

A) Smart Spectacle Hardware Module

- **Processing Unit:** Raspberry Pi Zero 2 W (edge computing)
- **Vision Sensor:** Raspberry Pi Camera Module V2 (real-time video input)
- **Distance Sensor:** VL53L1X LiDAR sensor (distance and depth measurements)
- **Motion Sensor:** MPU6050 IMU (head orientation + movement)
- **User Controls:** Push button (ON/OFF)
- **Storage:** MicroSD card (OS + model + logs)
- **Power Unit:** LiPo battery / power bank (portable power)
- **Audio Output:** Earphone (warnings + guidance)
- **Frame & Wiring:** Sunglass frame + connectors

B) Software/Processing Layer

- Image capture + preprocessing
- Lightweight object detection model for edge-based processing
- Distance-based hazard estimation (LiDAR)
- Motion/orientation handling (IMU)
- Sensor fusion + decision logic
- Audio feedback generation

4.3 High-Level System Diagram (Components)

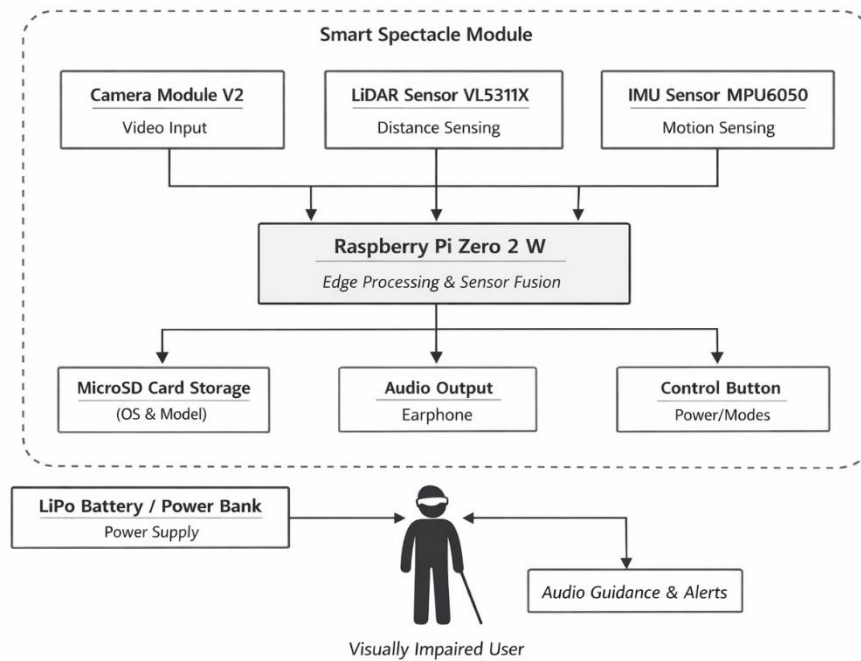


Figure 1 : High-Level System Architecture

4.4 Highlight: Individual Component and Its Parts (Smart Spectacle Hardware Module)

The Smart Spectacle Hardware Module can be divided into three main functional sub-units:

1. Sensing Unit

This unit is responsible for collecting environmental and motion data.

- **Camera Module V2** - captures the visual scene for object detection.
- **VL53L1X LiDAR Sensor** - measures distance and detects obstacles or elevation changes.
- **MPU6050 IMU Sensor** - detects head orientation and motion.

2. Edge Processing Unit

This unit performs real-time data processing and decision making.

- **Raspberry Pi Zero 2 W** - acts as the main edge computing device.
- **Local inference (object detection)** - processes camera images using AI models.
- **Sensor fusion and hazard decision** - combines data from multiple sensors to detect obstacles and hazards.

3. Feedback and Control Unit

This unit provides alerts to the user and controls system operation.

- **Earphone** – provides audio alerts and navigation guidance.
- **Push button** – used for power control or system mode selection.
- **Power system (battery management)** – supplies power to the wearable device.

4.5 Component Interactions

- **Camera → Pi:** Sends video frames for computer vision.
- **LiDAR → Pi:** Sends distance readings used to estimate how far obstacles/stairs are.
- **IMU → Pi:** Sends motion/orientation data to support stability and improve decisions (e.g., reduce false alerts while moving head).
- **Pi → Earphone:** Sends warnings/instructions (e.g., “Obstacle 1.5 meters ahead”).
- **Battery → All:** Powers the system; Pi manages safe shutdown if low power

4.6 Data Flow Explanation

- 1. Capture:** Camera captures frames continuously.
- 2. Sense distance:** LiDAR measures distance in front/target direction.
- 3. Sense motion:** IMU provides head movement and orientation.
- 4. Process on edge:** Raspberry Pi runs:
 - image preprocessing
 - object detection
 - hazard classification (obstacle/staircase/wall)
 - sensor fusion (combine vision + distance + motion)
- 5. Decision:** Determine hazard type + distance + urgency.
- 6. Feedback:** Generate real-time audio alert to earphone.

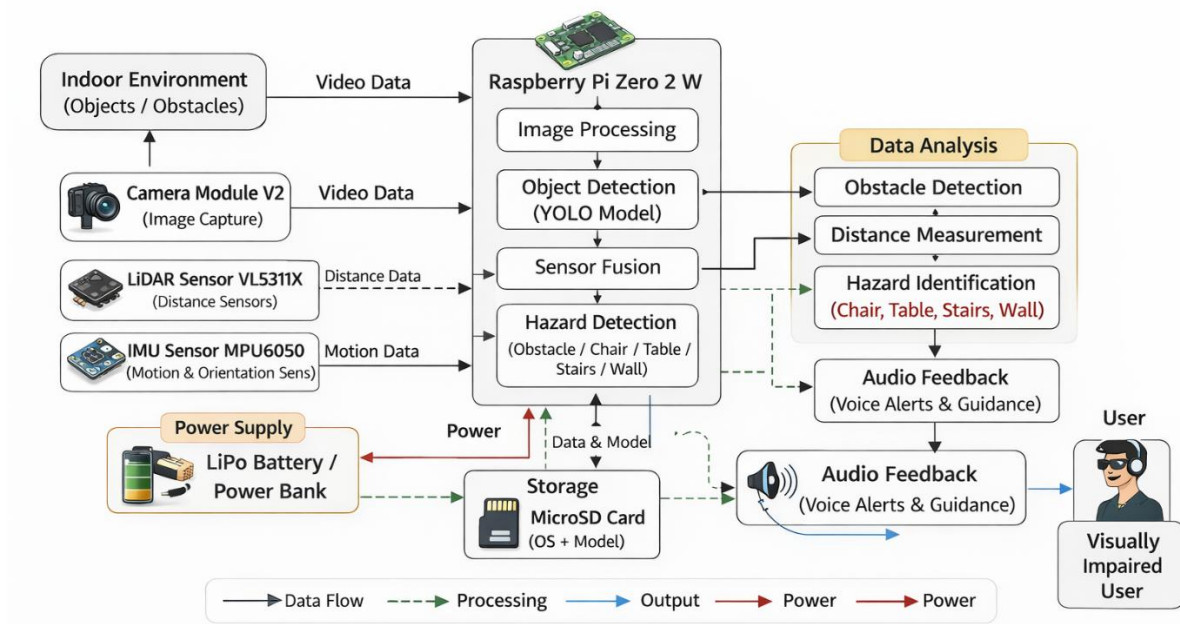


Figure 2 : Data Flow of the Proposed Smart Spectacle System

4.7 Scalability and Security Considerations

The proposed smart spectacle system is designed with scalability in mind so that additional assistive features can be integrated in the future without requiring major modifications to the system architecture. Since the system uses a modular hardware design and edge computing platform, new functionalities can be added to enhance navigation assistance for visually impaired individuals.

Scalability

Possible future scalability improvements include:

- **Product recognition in shops and supermarkets:** The system can also be improved to provide the recognition of products in stores and shopping malls, along with the audio information about the products, including the name of the products, the price of the products, and the expiry date of the products. This will help the visually impaired individuals during their shopping activities.
- **Outdoor navigation support:** The system can also be improved to provide support to the visually impaired individuals during their outdoor activities using other technologies, like GPS.
- **Known person recognition:** The system can also be improved to provide the recognition of known persons within the indoor environment, including shopping malls and the workplace. The system will provide an audio notification when a known person is recognized.

Security Considerations

Security and privacy are important considerations in the development of assistive wearable devices.

- **Privacy protection:** The system will focus only on detecting environmental objects and obstacles. Any captured images will not store personal identity information.
- **Offline processing:** The proposed system primarily performs data processing on the Raspberry Pi edge computing platform. This reduces the need to transmit sensitive data over the internet.
- **Secure data handling:** Any data collected during the test will only be used for research purposes and will not include any information that will identify individuals.
- **User safety:** The system will provide clear audio alerts to ensure that visually impaired users receive reliable navigation assistance while maintaining safe operation.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The smart spectacle system has various stakeholders who will be impacted by the system.

- **Visually Impaired Users:** These are the main users of the system. They will utilize the smart spectacle system to assist with navigation in indoor environments and detect obstacles within the environment.
- **Researchers and Developers:** The researchers and developers of the proposed smart spectacle system will be the primary stakeholders, as they will design, develop, and test the system.
- **Caregivers and Family Members:** The caregivers and family members of visually impaired individuals will also be stakeholders of the proposed smart spectacle system, as they will provide assistance to visually impaired persons.
- **Assistive Technology Organizations:** Organizations that promote accessibility technologies and may adopt the proposed system in the future.

5.2 Functional Requirements

The functional requirements outline what functions must be performed by the system.

FR1: The system should be able to capture real-time visual data through the camera module.

FR2: The system should be able to detect obstacles and objects indoors through computer vision techniques.

FR3: The system should be able to determine the distance from obstacles through the LiDAR sensor.

FR4: The system shall detect user movement and head orientation using the IMU sensor.

FR5: The system shall process the sensor data in real time using the Raspberry Pi edge computing platform.

FR6: The system shall provide audio alerts to the user through an earphone when obstacles or hazards are detected.

FR7: The system shall operate primarily in offline mode without requiring continuous internet connectivity.

FR8: The system shall provide the user with the ability to power the system on and off using a push button.

Smart Spectacle System – Use Case Diagram

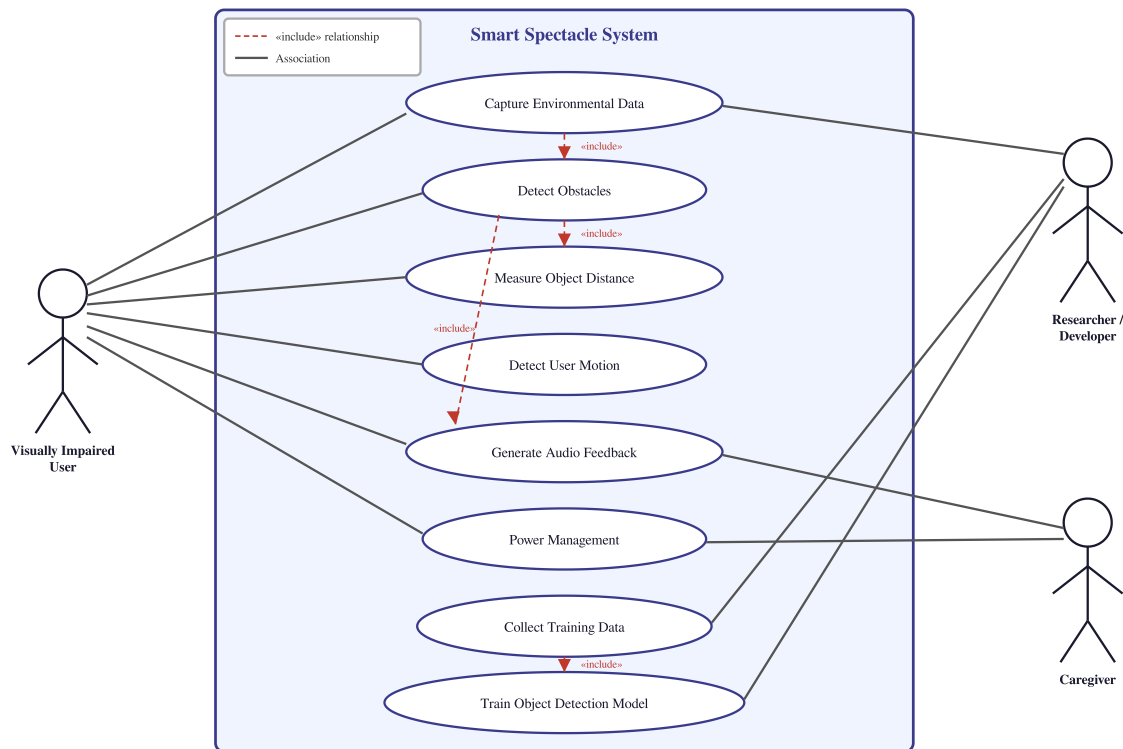


Figure 3 : Use Case Diagram

5.3 Non-Functional Requirements

The non-functional requirements define the quality attributes of the system.

- **Performance:** The system should process sensor data and provide obstacle detection feedback in real time with minimal delay.
- **Usability:** The smart spectacle should be lightweight, comfortable, and easy to use for visually impaired individuals.
- **Reliability:** The system should consistently detect obstacles and hazards in indoor environments.
- **Security and Privacy:** The system should ensure that captured data does not store personal or sensitive information.
- **Portability:** The device should be compact and wearable so that users can easily carry it during daily activities.
- **Energy Efficiency:** The system should consume low power to allow longer operating time with the battery.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed smart spectacle system will be visually impaired persons who need assistance with indoor navigation and obstacle detection. The proposed smart spectacle system will be very useful for persons who need assistance with movement in indoor places such as homes, offices, learning institutions, and other buildings.

Potential customers include:

- Visually impaired persons who need affordable assistive navigation technology
- Rehabilitation centers and hospitals for visually impaired persons
- Assistive technology organizations and accessibility programs
- Educational institutions with visually impaired persons

Customer Persona: The primary customer persona for the proposed smart spectacle system will be visually impaired persons who need assistance with indoor navigation and who need a lightweight, wearable, and affordable assistive device for their movement and navigation.

6.2 Value Proposition

User Problem	Proposed Solution	Value to the User
Visually impaired individuals face difficulty navigating indoor environments safely.	Smart spectacle with camera, LiDAR, and IMU sensors for obstacle detection.	Improves safe navigation and reduces risk of collisions.
Traditional tools such as white cane cannot detect overhead obstacles or staircases.	Multi-sensor integration for detecting objects and elevation changes.	Provides more reliable hazard detection.
Many assistive navigation systems depend on internet connectivity.	Edge computing using Raspberry Pi for offline processing.	Allows reliable operation without internet access.
Commercial smart assistive devices are often expensive.	Low-cost hardware components used in the proposed system.	Provides an affordable assistive solution for visually impaired individuals.

Table 1: Value Proposition of the Proposed Smart Spectacle System

6.3 Revenue Model

The proposed system can follow a product-based revenue model, where the smart spectacle device is sold directly to customers or through assistive technology distributors.

Possible revenue sources include:

- Direct sales of the smart spectacle device
- Partnership with hospitals and rehab centers
- Government or accessibility program support
- Licensing the technology to assistive device manufacturers

6.4 Cost Estimation

Component	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	8,890
Camera Module V2	8,000
LiDAR Sensor	1,550
IMU Sensor	680
Power System	1,700
Other Components	5,000
Estimated Prototype Cost	~ 25,820 LKR

Table 2 : Estimated Prototype Cost of the Proposed Smart Spectacle System

6.5 Pricing Strategy

The product pricing strategy will focus on affordability and accessibility.

Possible pricing approaches include:

- Affordable pricing model targeting developing countries
- Subsidized pricing through government or healthcare programs
- Partnering with NGOs that support visually impaired individuals

The retail price of the device will be around 30,000 - 35,000 LKR.

6.6 Competitive Advantage

The proposed system has several advantages compared to existing assistive navigation technologies.

- Integration of **multiple sensors for improved obstacle detection**
- **Edge-based processing** allowing operation without internet connectivity
- **Compact wearable design** suitable for smart spectacles
- Lower cost compared to commercial assistive smart glasses

6.7 Intellectual Property Considerations

The proposed system may include novel design elements related to **multi-sensor integration and wearable assistive navigation technology**.

Potential intellectual property considerations include:

- Patent protection for the system design and sensor integration method
- Copyright protection for the software algorithms used in the system
- Possible collaboration with technology organizations for commercialization

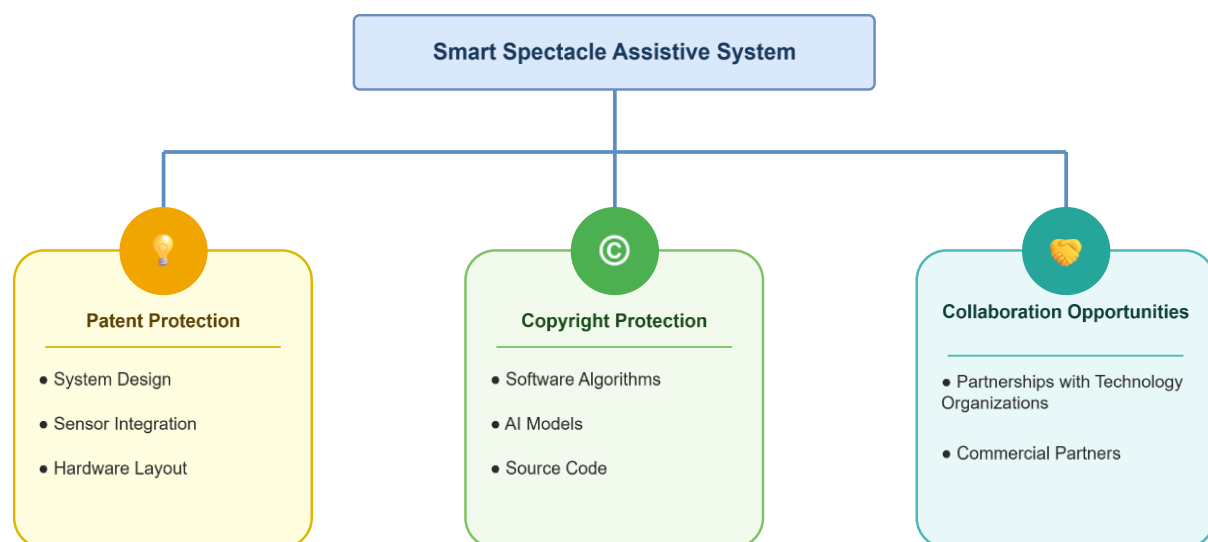


Figure 4 : Intellectual Property Protection Framework

7. BUDGET AND JUSTIFICATION

7.1 Estimated Budget Breakdown

The estimated budget required to develop the prototype of the proposed **IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired** is presented in Table 3.

Component	Quantity	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	1	8,890
Raspberry Pi Camera Module V2	1	7,950
VL53L1X LiDAR Sensor	1	1,550
MPU6050 IMU Sensor	1	680
Li-Po Battery (3.7V 3000mAh)	1	1,550
MT3608 Boost Converter	1	150
MicroSD Card (32GB)	1	3,100
Earphone	1	810
Sunglass Frame and Wiring	1	1,000
Estimated Prototype Cost		~ 25,680 LKR

Table 3 : Estimated Budget Breakdown for Prototype Development

7.2 Hardware Cost Justification

The bulk of the budget is spent on hardware components that are necessary for the building of the prototype system.

- **Raspberry Pi Zero 2 W:** This will be used for the building of the embedded processing unit of the object detection system.
- **Raspberry Pi Camera Module V2:** This component will be used for the collection of data for the building of the project.
- **VL53L1X LiDAR Sensor:** This sensor will be used for the detection of distance for the detection of obstacles such as staircases.

- **MPU6050 IMU Sensor:** This sensor will be used for the detection of the orientation of the head, thus improving the accuracy of the navigation.
- **Li-Po Battery & Boost Converter:** This will be used to power the portable object detection system. It will also be used to power the raspberry pi.
- **MicroSD Card:** This will be used to store the operating system, software programs, and the machine learning model used by the object detection system.
- **Earphone:** This will be used for the provision of audio feedback to the user of the object detection system.
- **Sunglass Frame & Wiring:** This will be used for the connection of all the hardware components to the object detection system.

7.3 Software Requirements

The proposed system primarily utilizes **open-source software tools and libraries**, which significantly reduce development and deployment costs. These tools support computer vision, sensor integration, and system control for the smart navigation device.

Software Tool	License Cost
Python Programming Language	Free
OpenCV Library	Free
Raspberry Pi OS	Free
TensorFlow Lite (for AI processing)	Free

Table 4 : Software Tools Used in the Proposed System

7.4 Cloud Services

The proposed system is designed to operate mainly using **edge computing**, where most data processing occurs locally on the Raspberry Pi device.

This design reduces dependence on cloud computing services and avoids additional operational costs. Therefore, no cloud service expenses are required for the prototype development stage.

7.5 Human Effort Cost

The development of the system will be carried out by the **research team members as part of their academic research project**. Therefore, no direct financial cost is allocated for human resources during the prototype development stage.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

Member	System Component	Key Tasks
Member 1	Intelligent Vision-Based Danger Detector 	Task 1: Collect and prepare image dataset for indoor hazards. Task 2: Configure YOLO object detection model for edge devices. Task 3: Train and optimize the model using labeled datasets. Task 4: Evaluate detection accuracy and performance. Task 5: Integrate detection system with the smart spectacle platform.
Member 2	Intelligent Security & Remote Guardian System 	Task 1: Design and develop the guardian mobile application. Task 2: Implement fall detection and inactivity monitoring algorithms. Task 3: Develop geo-fencing functionality for safe zones. Task 4: Integrate cloud database for real-time monitoring. Task 5: Implement SOS emergency alert system.
Member 3 (ME)	Smart Spectacle Hardware Module with Multi-Sensory Integration 	Task 1: Design the wearable smart spectacle hardware structure. Task 2: Integrate Raspberry Pi with camera, LiDAR, and IMU sensors. Task 3: Implement sensor data acquisition and synchronization. Task 4: Develop audio output interface for navigation alerts.

		Task 5: Implement push-button control for system interaction.
Member 4	Hybrid Indoor Positioning System with Offline Capability 	Task 1: Develop indoor localization algorithm using IMU sensors. Task 2: Implement vision-based landmark recognition. Task 3: Integrate dead-reckoning for indoor positioning. Task 4: Combine vision and IMU data for hybrid localization. Task 5: Integrate positioning system with navigation and hazard detection modules.

Table 5 : Task Breakdown per Member for the Proposed Smart Spectacle System

8.2 Timeline Alignment

The development process of the proposed IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired will be carried out in an aligned manner with an appropriate timeline so that it can be achieved systematically during the entire research period. The project will be achieved through different key milestones that will be included in the research process.

The project will be aligned with the following timeline:

- Phase 1 – Literature Review and Problem Analysis (Month 1):**
Thorough research will be performed based on the existing research works on assistive technologies, indoor navigation systems, computer vision models, and IoT-based wearable devices.
- Phase 2 – System Architecture and Design (Month 2):**
Design of the entire system, including hardware as well as software components, shall be conducted.
- Phase 3 – Hardware and Software Development (Month 3-4):**
The development of the individual components of the entire system will be performed.

- **Phase 4 – System Integration and Testing (Month 5):**
All the components of the entire system will be integrated and the testing of the entire system will be performed.
- **Phase 5 – Evaluation and Documentation (Month 6):**
Analysis of the entire system, completion of the research report, and completion of the project's presentation.

8.3 Balanced Workload Distribution

The project workload is distributed among the four members of the group based on the technical expertise of each member and the needs of the proposed project. Each member of the group is responsible for the development of a particular module of the project, including vision-based hazard detection, the remote guardian monitoring system, the smart spectacle hardware module, and the hybrid indoor positioning system. This ensures the development of all the major parts of the project efficiently, with all the members of the group participating equally.

Besides the development of the project's individual parts, some activities such as integration, testing, and report writing will be undertaken jointly by all the members of the group to ensure the equal participation of all members in the successful completion of the research project.

9. GANTT CHART

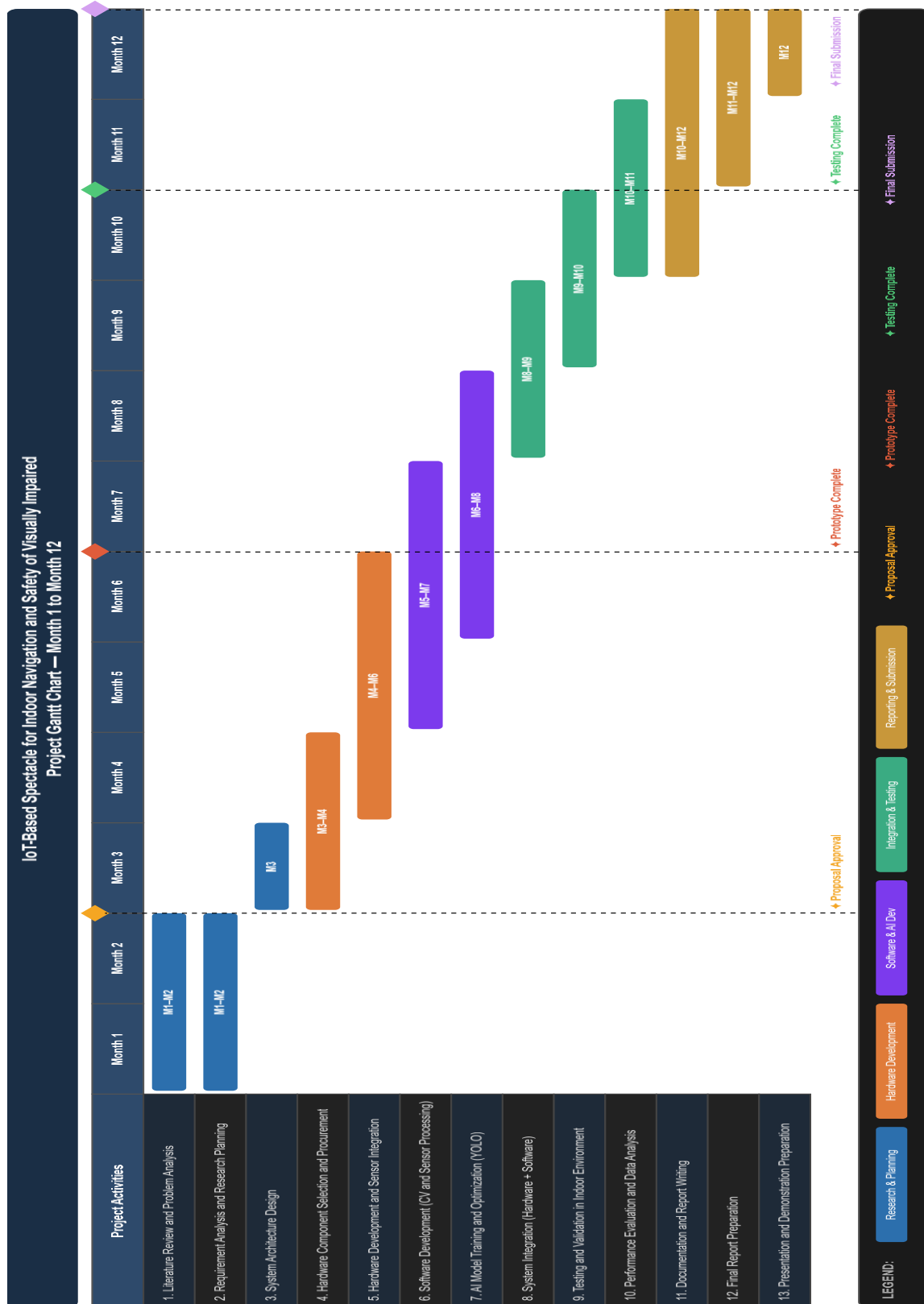


Figure 5 : Gantt Chart for the Proposed Smart Spectacle Research Project

10. REFERENCES LIST

- [1] World Health Organization, "Blindness and vision impairment," World Health Organization, Geneva, 2023.
- [2] R. Velázquez, "Wearable Assistive Devices for the Blind," in *Wearable and Autonomous Biomedical Devices and Systems for Smart Environment: Issues and Characterization*, vol. LNEE 75, Berlin, Springer, 2010, pp. 331-349.
- [3] D. Plikynas, A. Žvironas, A. Budrionis and M. Gudauskis, "Indoor Navigation Systems for Visually Impaired Persons: Mapping the Features of Existing Technologies to User Needs," *Sensors*, vol. 20, no. 3, p. 1–32, 2020.
- [4] K. Swathi, P. Gayathri, D. Manasa, G. Pavani and B. Likhitha, "Smart Glasses for Visually Impaired People Using IoT," *International Research Journal on Advanced Engineering and Management (IRJAEM)*, vol. 2, no. 5, pp. 1390-1394, 2024.
- [5] S. G. S. B. C. O. A. M. S. L. a. J. F. Rosemary O’Keeffe, "Long Range LiDAR Characterisation for Obstacle Detection for use by the Visually Impaired and Blind," in *2018 IEEE 68th Electronic Components and Technology Conference (ECTC)*, San Diego, CA, USA, 2018.
- [6] J.-Y. Lin, C.-C. Yao, C.-L. Chiang, M.-C. Chen and M.-J. Wu, "Smart Glasses Application System for Visually Impaired People Based on Deep Learning," in *2020 Indo – Taiwan 2nd International Conference on Computing, Analytics and Networks (Indo-Taiwan ICAN)*, Taitung, Taiwan, 2020.
- [7] Z. Chen, X. Liu, M. Kojima, Q. Huang and T. Arai, "A Wearable Navigation Device for Visually Impaired People Based on the Real-Time Semantic Visual SLAM System," *Sensors*, vol. 21, no. 4, pp. 1-19, 2021.
- [8] Y. Chen, J. Shen and H. Sawada, "A wearable assistive system for the visually impaired using object detection, distance measurement and tactile presentation," *Intelligence & Robotics*, vol. 3, p. 420–435, 2023.
- [9] D. López-de-Ipiña, T. Lorigo and U. López, "Indoor Navigation and Product Recognition for Blind People Assisted Shopping," in *Proceedings of the 3rd International Workshop on Ambient Assisted Living (IWAAL 2011)*, Málaga, Spain, 2011.

- [10] D. V. Kumar, K. I. Priyadarsini, V. R. Kammampati and E. P. Konakalla, "Real-Time Object Recognition with Voice Feedback for Visually Impaired Based on Raspberry Pi," in *2024 2nd International Conference on Device Intelligence, Computing and Communication Technologies (DICCT)*, 2024.
- [11] J. Redmon, S. Divvala, R. Girshick and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016.

11. APPENDICES

Appendix A - Survey Instruments

As a follow-up, the following is a sample questionnaire that seeks to understand the challenges faced by visually impaired individuals when moving around within an indoor environment. Additionally, the questionnaire seeks to find out the possible application of wearable assistive technology devices.

Target Audience: The target audience for the survey will be individuals who are visually impaired, caregivers of individuals who are visually impaired, and professionals who are knowledgeable about assistive technologies.

Survey Questions

1. What type of difficulties do visually impaired individuals commonly face when navigating indoor environments?
 - Difficulty identifying obstacles
 - Difficulty recognizing stairs or elevation changes
 - Difficulty determining direction or location
 - All of the above
2. Which assistive device is commonly used by visually impaired individuals for mobility?
 - White cane
 - Guide dog
 - Smartphone navigation applications
 - Wearable assistive devices
3. How important is obstacle detection for safe indoor navigation?
 - Very important
 - Important
 - Moderately important
 - Not important

4. What type of feedback would be most useful for a smart navigation system?
 - Audio instructions through earphones
 - Vibration alerts
 - Visual notifications
 - Combination of audio and vibration

5. How useful would a wearable smart spectacle device be for assisting visually impaired individuals in indoor navigation?
 - Very useful
 - Useful
 - Slightly useful
 - Not useful

6. Which feature would be most important in a smart navigation device for visually impaired users?
 - Obstacle detection
 - Staircase detection
 - Real-time navigation assistance
 - Emergency alert system

Appendix B - Detailed System Diagrams

• Smart spectacle hardware architecture diagram

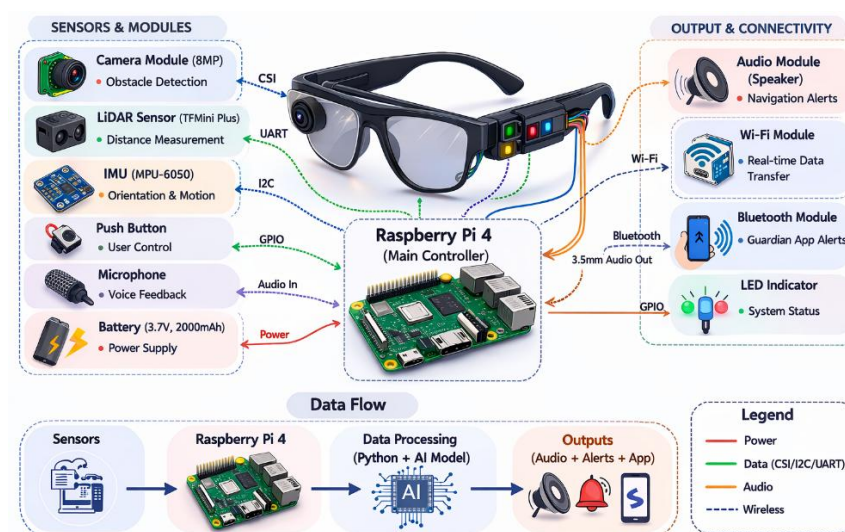


Figure 6 : Smart Spectacle Hardware Architecture Diagram

- **Data processing pipeline**

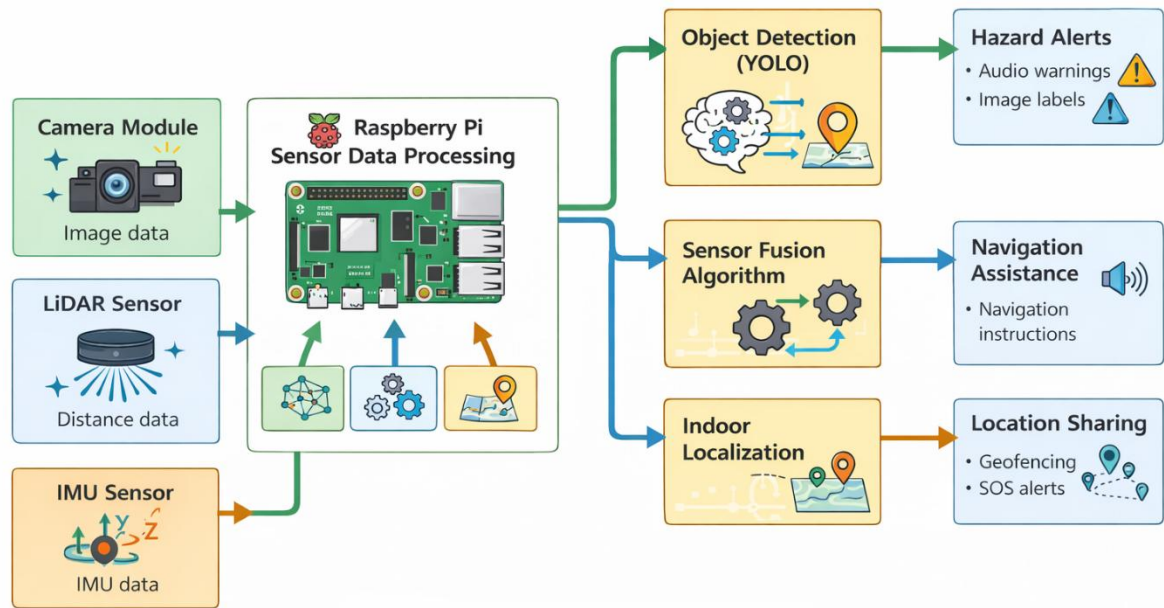


Figure 7 : Sensor Data Processing Pipeline

Appendix C – Preliminary UI Mockups

This appendix presents the preliminary user interface designs for the Guardian Monitoring Mobile Application, which allows guardians to monitor the safety and location of visually impaired users using the proposed smart spectacle system.

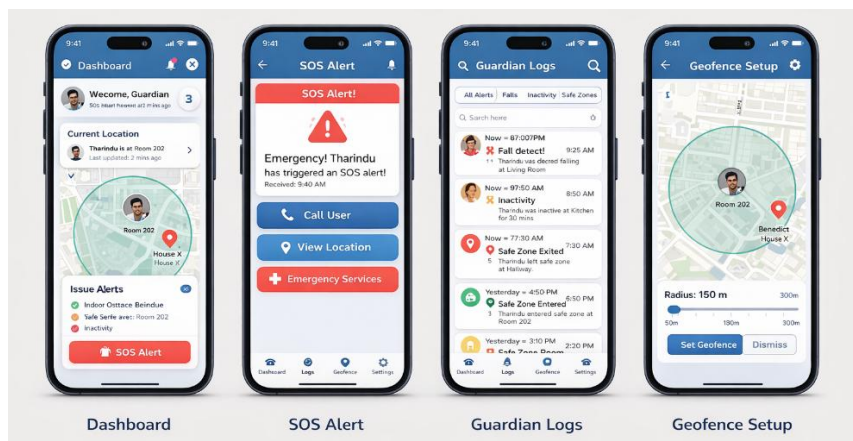


Figure 8 : Preliminary User Interface Mockups for the Guardian Monitoring Mobile Application

Appendix D – Risk Analysis

Risk Factor	Impact	Probability	Mitigation Strategy
Hardware component failure (Raspberry Pi or sensors)	The system may stop functioning or produce incorrect results	Medium	Use reliable hardware components and perform regular hardware testing
Sensor inaccuracies (LiDAR or IMU errors)	Incorrect obstacle detection or navigation guidance	Medium	Implement sensor fusion and calibration techniques
Camera performance affected by lighting conditions	Object detection accuracy may decrease in low-light environments	Medium	Combine camera detection with LiDAR distance sensing
Battery power limitations	Device may shut down during operation	Medium	Use high-capacity Li-Po battery and optimize power consumption
Overheating of embedded device	Reduced system performance or hardware damage	Low	Provide proper ventilation and optimize processing workload
Loose wiring or hardware integration issues	Sensors may stop communicating with the processing unit	Low	Secure wiring and perform hardware stability testing

Table 6 : Risk Analysis for the Smart Spectacle Hardware Module (Component 3)

Appendix E – Ethical Considerations

This research is carried out following the guidelines on research ethics provided by the Sri Lanka Institute of Information Technology. This institute has an Ethics Review Committee (ERC), which deals with the review of the research proposal and ensures that all the research activities are carried out ethically.

The following ethical principles are considered in this research project:

- **Personal Data:**
This research does not require any personal data from any individual. If any data is required in the experiment for the system, it is related to objects in the environment, such as furniture, doors, staircase, etc.
- **Privacy protection:**
If images are captured during experiments in indoor environments, they will not intentionally include identifiable individuals. Any accidental appearance of people will not be stored or used for analysis.
- **Responsible use of technology:**
The developed smart spectacle system is only intended to assist and enhance the safety and mobility of people with visual impairments.
- **Compliance with institutional ethics guidelines:**
This research has been conducted based on the ethical guidelines suggested by the SLIIT ERC to carry out ethical academic research.
- **Academic integrity:**
All the information sources, data sets, and research material used in the current study are properly credited to ensure academic integrity.

SLIIT Ethics Review Committee:

<https://www.sliit.lk/about/about-sliit/ethics-review-committee>